CHEMISTRY'S VICTORY OVER DISEASE

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The Koch treatment is primarily a treatment for incurables and it has been successful. The facts cannot be denied. This does not mean that everyone can be snatched from the grave. It means that a new advance in the conception of disease processes has been worked out and that the new details have been carefully analyzed and mastered and put to use in securing accurate protection.

With the knowledge that the poisons that cause disease work through their photochemic properties, fluorescence, and negative oxidation catalysis, as Dr. Koch has definitely set forth; the next step was not difficult. It is a matter of turning the trick of the enemy against itself. How this is done is interesting indeed.

HOW DISEASE IS PRODUCED

The complex organic compounds that cause disease are either produced directly by bacteria or exist as hidden complexes in the protein of different species. We may say that all species are chemically poisonous to each other but in the case of pathogenic bacteria, the poisonous principles are free or easily set free within the body of another species; while in the so-called innocent species, be they bacteria or plants or animals eaten as food, the toxic elements are tightly bound and not set free by the digestive processes in the stomach and intestines, but are destroyed thereby. These toxic elements are held "deep" within the protein structure and can be set free by parental digestion, that is, by digestion of the type that goes on within the bloodstream or in the tissue cells themselves. This type of digestion is the kind that breaks down the worn-out tissue cells to make way for new cells. However, when one's own tissues are digested in this way, no toxic effect of noticeable magnitude is experienced, but when large amounts of one's tissues have to be digested and eliminated, such as after extensive injuries and burns, there may be sufficient toxic material introduced into the blood stream to produce severe sickness or death. One's own tissues contain toxic elements, therefore. Dr. Koch isolated and identified some of them in 1912, 1913, and 1914, and published his findings in several of the "*Journal of Biological Chemistry*" and the "*Journal of Laboratory and Clinical Medicine,*" published at that time.

Injured blood and tissues are first coagulated before being digested and so the speed of their chemical disintegration and absorption is safely reduced under ordinary circumstances. When a foreign protein from some other species enters the bloodstream or is generated within the tissues by bacteria harbored there,

this protein is "adsorbed" by the tissues of the host that have first contact and held that way as a part of the protein of the tissue. It may not cause any disturbance, especially if the tissues have had no experience with it or are able to destroy it. But if they have had former contact, or have further contact in ten days to a few weeks later, they learn to digest the foreign protein and tear it down and in so doing set free its toxic elements at a dangerous rate.

There is a colloidal phase to this affair, which does not involve the breaking down of the foreign protein, but involves a coagulation, or pseudo-coagulation, or precipitation of the foreign protein, or of the host's protein, or both. These proteins may be lysed or split up physically so they are not dispersed in a normal way and gel or flocculate and clog the circulation in a disease or death-producing way. Fundamentally it is a matter of removing the electric charges from the tissue or blood colloids and paralyzing the production of the energy that supports dispersion.

No matter in which way the disease-producing effect is accomplished, it would not do so if the toxic elements were destroyed. Here is where Dr. Koch's work includes a major victory, for he identified both the toxic action and the colloidal injury, as the results of the free valency activities of certain unsaturated residues of the toxic molecule. They both are photochemic effects, namely fluorescence of a specific grade and oxidation quenching or negative oxidation catalysis.

HOW THE CAUSE OF DISEASE IS OVERCOME

Dr. Koch decided that the thing to do was to saturate these free valencies in a way that is natural within the body and, therefore, harmless. By saturating these valencies with oxygen, they lose their toxic action and are neither fluorescent anymore nor can they interfere with the oxidations of the tissues through which the energy of charging the colloids and the energy of function is produced.

We said a moment ago that Dr. Koch's Therapy turned the trick of the enemy against itself. This is a simple matter to understand now, but in 1914, when Dr. Koch was struggling with the problem, it was shrouded in darkness. Indeed, the science of photochemistry was only being founded, and it is still in its infancy. Since this science is the mathematics of chemical energy transmission and has to do with electronic phenomena that are difficult to manipulate even today. We marvel at the fact that Dr. Koch could put into use such phenomena as were misunderstood grossly at that time and whose laws are just becoming accurately formulated today. He has illustrated the extremely interesting phenomena very nicely as follows:

We may look upon the valencies or electric charges that hold atoms together as arms or electric magnets. Each atom has a definite number of them. Carbon

has four, oxygen two, hydrogen one etc. in any compound where only two or three of the valencies of carbon are occupied and the others free, the latter are under the influence of the electromagnetic field, and will vibrate in accordance with their ability to do so under the conditions of this electromagnetic field. It is much like the changes in the radio wave under conditions of static. The free valencies can vibrate under influence of energy coming to them and may do so in a way whereby they store up energy. But since they cannot hold it but momentarily, they again send it forth by their own vibrations. In doing so they degrade the energy they receive and issue it at a lower energy value. So they serve as receiving and as broadcasting stations. The shift in energy value can be just enough to make it specific for influencing certain chemical processes going on in the medium where it is held adsorbed or in intimate solution. The energy taken up by the free valencies can be either some radiation of special wavelength or the energy evolved in some chemical reaction. And it is either set free as a lower grade radiation or if some chemical system is present that absorbs energy of the same wavelength as the substance emits, then the energy can be accepted by this chemical system and pass into its chemical reactions giving them added impetus. This is the phenomenon of fluorescence that was so misunderstood when Dr. Koch made efficient us of it and decided that it interfered with normal energy transfer in the tissue cells in two ways.

The first is the interruption of the normal energy transmission through change to a degraded form, the reduction to a wavelength of lower energy value and vibration rate. The other is the transfer of the degraded energy to the chemical system of some functional unit that is able to accept the energy at the reduced vibration rate and thus receive energy that passes into its chemical processes and keeps them going. Thus the contractions of the muscles in asthma, and the increased secretions of the mucous membranes in asthma and hay fever are maintained. In the same way the mechanism that conducts cell division is kept going so the cells can multiply in a forced way to form cancer growths and other tumors. The phenomena of insanity and neuritis are explainable on the same basis, but here the conductile elements of nerve cells associated in some act or concept carry the impulse continuously and produce fixed ideas, delusions, etc.

Dr. Koch ascertained also that fluorescent structures could quite effectively quench their own activity as well as the activity of other free valencies of similar type. Thus, if the structure of the rest of the molecule was sufficiently large and stable to make the free valency sluggish, they may absorb the energy of other free valency activity very effectively and obliterate it. In other words, the free valency activity of the catalysts that conduct the cell oxidations can be so absorbed that it is prevented from accomplishing its regular work. Thus the oxidations that are necessary for cell function and for burning disease-producing poisons, themselves, are wiped out or quenched.

Since similar atoms influence similar atoms and similar atomic groups influence similar atomic groups electromagnetically, the effects produced will depend upon which atom or group possesses the advantage, that is, the effect depends upon which group has the upper hand. In the production of cancer, for instance, the carcinogenic poison's free carbon valencies and oxygen valencies of the ethylene and carbonyl groups and the free nitrogen valencies of the imide group absorb the energy of the oxidation catalyst's valency activity and quench the oxidations as well as shift energy into cell reproduction activities.

In order to reverse the process and hold the upper hand, it appeared necessary to introduce such energetic oxidation activation that the free valencies of the mischief-makers would have to combine with oxygen and be inactivated and burned in decisive fashion and thus be eliminated from the field. This is not difficult, as you will see.

It happens that free valencies have definite habits. They want to combine with or take hold of other free valencies. Carbon atoms like to combine with oxygen, and if they do not have the chance they will combine with other carbon atoms when they can. Thus, they will be oxidized or they will yield to polymerizing activity, which makes their molecules even more effective at oxidation quenching.

In order to set such polymerizing valencies free and to cause them all to vibrate at the rate, which disposes them to take up oxygen, all that is needed is to introduce a very active set of free carbon and oxygen valencies into the field. Dr. Koch has worked out several extremely unstable molecules that serve well in this regard. Each possesses carbon and oxygen in so active a state that when similar atoms and groups of the oxidation quenchers and fluorescers come into their field of influence, they vibrate in unison with these activators and take up oxygen too. Or we may say that the remedial molecules activate oxygen so vigorously that this oxygen combines the free valencies that have been doing the mischief. So indeed the activated free valencies of the mischief-maker also has its influence upon its neighbor of the same kind and activates it too. Thus the field is changed to one in which the vibrations inducing oxidation are propagated from molecule to molecule until each pathogenic unit is burned up and can no longer pervert the tissue chemistry.

Thus as Dr. Koch stated a quarter of a century ago, the cause of the disease is converted into the cure, in so stating, this reaction he has outlined is a mechanism of immunity which is far more perfect than the vague formulation of the great Ehrlich and indeed, in the light of recent facts, it appears that the Koch Explanation is more in accord with Nature's characteristic efficiency and

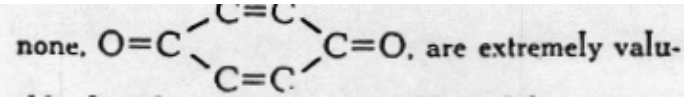
closer to the truth than the Elirlich Theory. The Koch philosophy has brought results in the vast field of incurable diseases, where Ehrlich's work had no prospects of results at all.

The scientific factors explained here are not matters of common knowledge among medical men. Probably not more than one percent of the hundred and fifty thousand physicians in North America have ever heard of photochemistry. Much less are they equipped to understand it. This is no discredit to them, however. But they should recognize their limitations and not comment on matters they know nothing about, even if people might expect them to have this knowledge. It is tragic that the editorial services of our chief medical magazines have been so stupid on this score.

NATURE OF THE PROTECTIVE AGENTS

The chemical structures that Dr. Koch has employed to secure the oxidation catalysis requisite for the destruction of pathogenic toxins are very simple indeed, and it is to this simplicity that the liability of their free valencies is due in large measure. The carbonyl group combined with ethylene and shared with ethylene groups present in the molecule, or the potential by virtue of properly placed hydrogen and hydroxyl, confers these powers. So definite is this property, that Dr. Koch has formulated a series of rules of chemical structure that must be met before a substance can exert protective or immunity powers.

The substances used most successfully are: Glyoxylide, O=C=C=O. This is the most unstable and the most reactive and beneficial of all. Malonide, O=C=C=C=O, and 1:4 Benzoquinone,



able but less active than Glyoxylide. Ketene, H(2)C=C=O, is also valuable, but comparatively slow in its action, and it may not be the victor over some extremely highly obstructive pathogenic molecule. Propargylic aldehyde has proven curative in certain cases also.

UNIVERSAL FIELD OF ACTION

Since the action of these structures is fundamental and destroys the very essence of pathogenic activity of disease producing agents, its held of action is wide indeed. In fact there is no disease theoretically or in practice that does not call for its use. Cancer has been the challenge to medicine ever since the dawn of history. Here is a cure. This is demonstrated in the "*Transactions of the Canadian Cancer Commission*" recently appointed by the Ontario Government. Actinomycosis is often incurable and confused with cancer. The

gumma stage of syphilis is often associated with cancer and confused with it. They have both been cured by this treatment after failing to respond to the usual methods. The degenerative diseases such as progressive muscular atrophy, multiple sclerosis, and the like have also made splendid recoveries.

Then there are the acute infections of fatal import; postoperative Staphylococcus meningitis, Streptococcus sore throat, pneumonia with serious kidney complications, coronary thrombosis, and others that bring the patient to death's door. They too depend upon the molecular structure discussed above to produce their injuries, and they too have been cured by use of a small dose or two of one of these extremely active molecules, after all other methods failed. The common cold with its sore throat and husky voice responds in a few minutes at times and is well in a few hours.

The common allergies that keep one miserable throughout life, hay fever, asthma, psoriasis, eczema, and so many others are effects of the chronic poisoning of the tissues, by the products of germ life imprisoned within the scars of focal infections. They too are overcome by the same oxidation catalysis. Since the life of such germs depends upon the poisons they liberate, they must die when these poisons are destroyed quicker than they can be produced. Thus the destruction of focal infections and their poisons rids the body completely of the cause of the allergy. It is not simple alleviation, but true cure. The clinical evidence indicate that the virus and protozoan infections depend upon toxins possessing similar atomic arrangements within their active groups to those produced by pathogenic bacteria, indeed the rapid recoveries from shingles, infantile paralysis, measles and mumps, demonstrate that they are even more vulnerable to the Koch Reagents than ordinary pathogenic germs. This seems to be true in cases of advanced syphilis and malaria also, and so the generalization is definitely warranted. To be able to place the toxins of the different pathogenic species in the same category of molecular arrangement, and to prove out one particular molecular arrangement that is specifically anti-toxic to each, is definitely an epochal advance in medicine.

One should not be surprised that the Koch system is not the usual method of treatment of any disease in vogue today. It is the Therapy of the future and will come into general use as rapidly as public and professional education creates a proper appreciation of its mechanism and merits.

Any physician of ordinary training who is in earnest in following the procedure conscientiously can use the method successfully.

It is now being used to save the lives of the incurables, after other methods fail. Soon it will be used to cure the sick before becoming 'incurable' and finally it will be used to prevent disease from successfully attacking the body at all.

Whereas physicians employing the method have been persecuted like the first users of diphtheria antitoxin, the day is close when it will be illegal not to use it; as is true of diphtheria antitoxin today. Progress in epochal movements has always been slow. The debris of entrenched greed is never easily or quickly dug out and destroyed. Still, in this instance, as ever in the past, time is replacing failure with success of the most practical kind, a victor over disease that is infinitely compatible with normal body chemistry. Indeed, the defect that invites disease can now be corrected. The missing protective principle can be restored. The problem is solved!

REFERENCES

Proceedings; Cancer Commission. Province of Ontario, Canada, Regarding the Koch
Treatment, 1938-1940, with Comment by D. H. Arnott. M.D.

Medical Record of New York, October 1920. *A New and Successful Treatment for Cancer*
Wm. F. Koch, Ph.D. M.D.

Cancer and Its Allied Diseases, 1927, 1929, 1933—Wm. F. Koch, Ph.D. M.D.

Natural Immunity, 1936, Wm. F. Koch, Ph.D. M.D.

The Chemistry of Natural Immunity, 1938, 1939, Wm. F. Koch, Ph.D. M.D.

Jour. of Biol. Chem., Vol. 12, p. 313, 1912, Wm. F. Koch.

Jour. of Biol. Chem., Vol. 15, pp. 43-61, 1913, Wm. F. Koch,

Journal of Laboratory and Clinical Medicine, Vol. 1, p. 299, 1916, Wm. F. Koch.

Journal of Medicine and Surgery, 1918, pp. 1-9, Wm. F. Koch.

C.R. de la Soc. *de Biol.,* Vol. 120, 1935, p. 106, Koch and Maisin.

Lectures, Koch, 1936, 1937, *The Chemical Structure that Determines Immunity to Disease.*